Machine Learning – CDS201

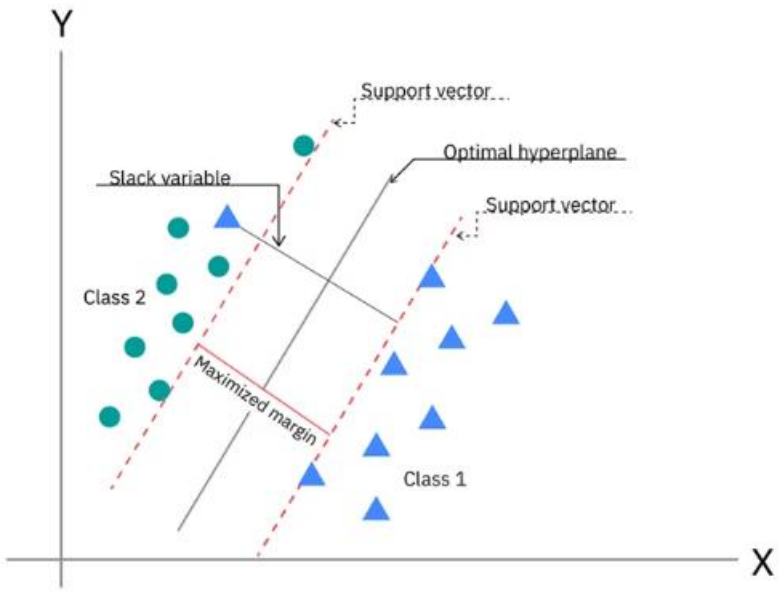
Module 3: Classification Models

Support Vector Machines (SVM)

- A support vector machine (SVM) is a supervised machine learning algorithm that classifies data by finding an optimal line or hyperplane that maximizes the distance between each class in an N-dimensional space.
- **Idea:** Distinguish between two classes by finding the optimal hyperplane that maximizes the margin between the closest data points of opposite classes.
- The number of features in the input data determine if the hyperplane is a line in a 2-D space or a plane in a n-dimensional space.
- Since multiple hyperplanes can be found to differentiate classes, **maximizing the margin** between points enables the algorithm to find the best decision boundary between classes.
- Enables it to generalize well to new data and make accurate classification predictions.
- The lines that are adjacent to the optimal hyperplane are known as **support vectors** as these vectors run through the data points that determine the maximal margin.
- Thus, used for both linearly and non-linearly separable problems.

Support Vector Machines (SVM)

- Kernel trick: For non-linearly separable, kernel functions are used to transform the data to a higher-dimensional space to enable linear separation.



Types of SVM classifiers

- Linear SVMs - linearly separable data

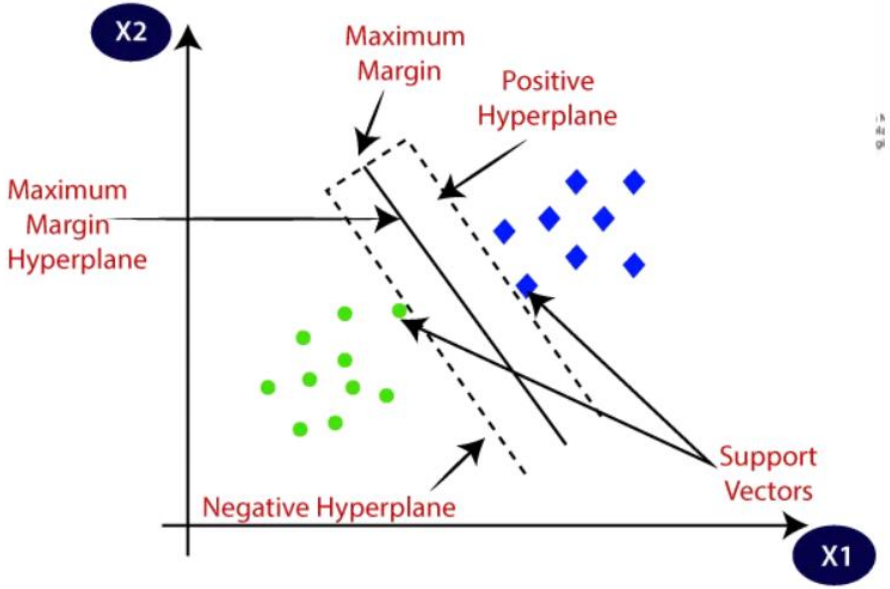
- Nonlinear SVMs - data in real-world scenarios are not linearly separable

Linear SVMs

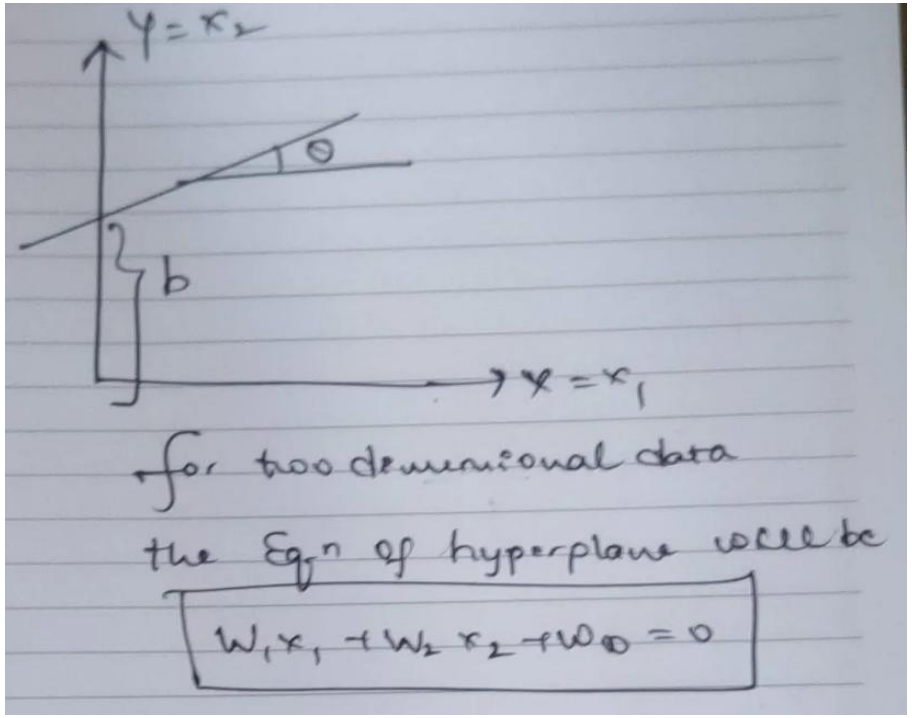
- Hard Margin
- Soft Margin

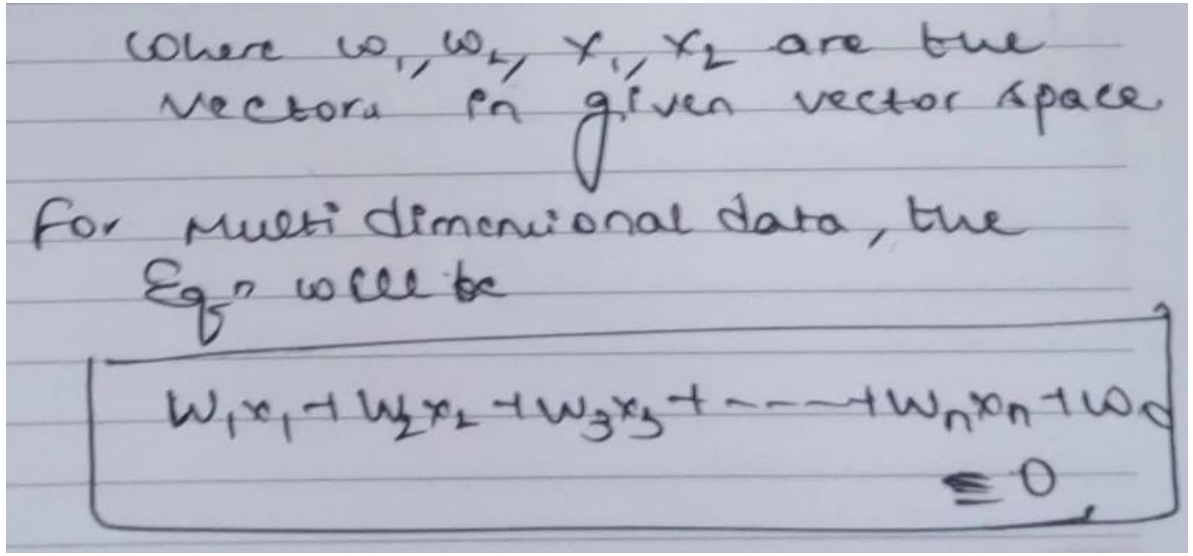
Hard margin – linear SVM

- Theme of this algorithm : Create a decision boundary that maximizes the margin between the classes in the data, based on the nearest data points the margin will be maximized, and these nearest data points are known as the **Support vectors.**
- **W**henever a new data point is witnessed, the new data point will be correctly classified because of that decision boundary that separates the classes with maximum margin - **Maximum Margin Classifier .**
- Despite the advantages, the hard margin classifier in the SVM is limited only to Linearly separable problems by forcing some constraints to satisfy.



Clarity on your line/plane equation – (repeated content)

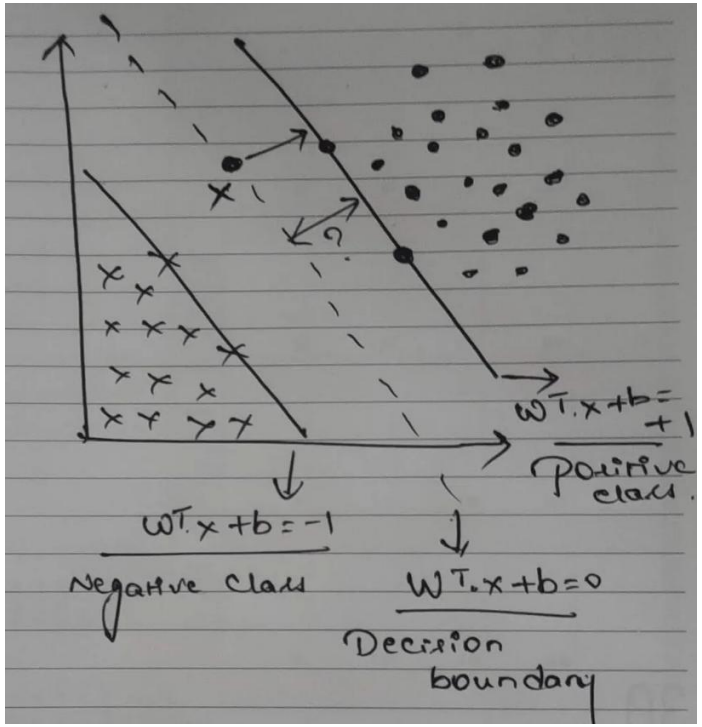




- If we observe carefully the equation is nothing but the dot product of two vectors W and X, so this equation can be further simplified as **W^T*X+W0=0.** where W^T represents the transpose of the W matrix which does dot product with X. W0 is the bias term, it becomes zero when it passes through the origin.

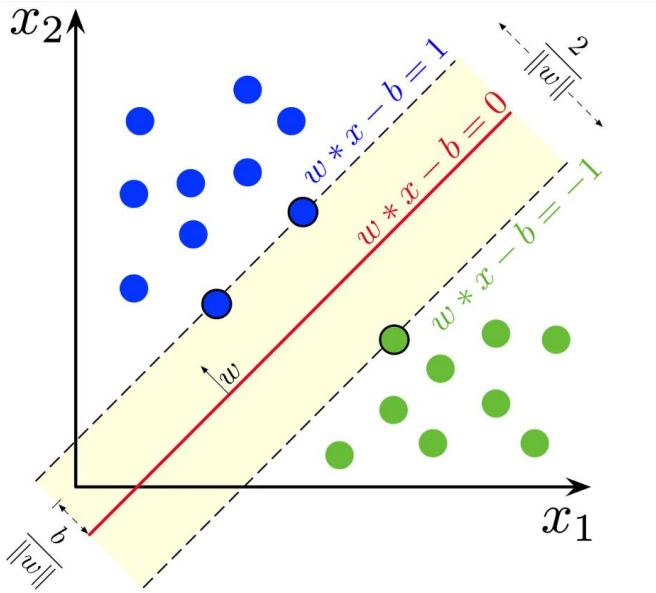
Hard margin classifier

- Take a linearly separable problem which has two classes in it classified by the Positive class and negative class.
- **for y=+1 -> W^T*X+b≥1**, the reason why it always should be greater than or equal to +1 is, whenever a data point is labeled as the positive class it might be a support vector where the dot product equals +1or it might be greater than +1.
- **for y=-1 -> W^T*X+b≤-1**, the reason why it always should be lesser than or equal to -1 is, whenever a data point is labeled as the negative class it might be a support vector where the dot product equals -1 or it might be lesser than -1.



Hard margin classifier

- Calculate the total distance of the margin (x2 – x1) x2 and x1 are the support vectors for class +1 and -1:



$$(-)\quad w^T x_1 \cancel{+ b} = (+)\,1$$
$$w^T x_2 \cancel{+ b} = +1$$

$$w^T (x_2 - x_1) = 2$$

$$\frac{w^T}{\|w\|}(x_2 - x_1) = \frac{2}{\|w\|}$$

$$x_2 - x_1 = \frac{2}{\|w\|} \Leftarrow \text{Max Margin}$$

$$x_2 - x_1 = \text{Min } \frac{\|w\|}{2} \quad (\text{Also})$$

S.T $\quad y_i(w^T \cdot x_i + b) \geq 1$

$\forall i = 1, \ldots, n$

for Support vectors

$$y_i(w^T \cdot x_i + b) = 1$$

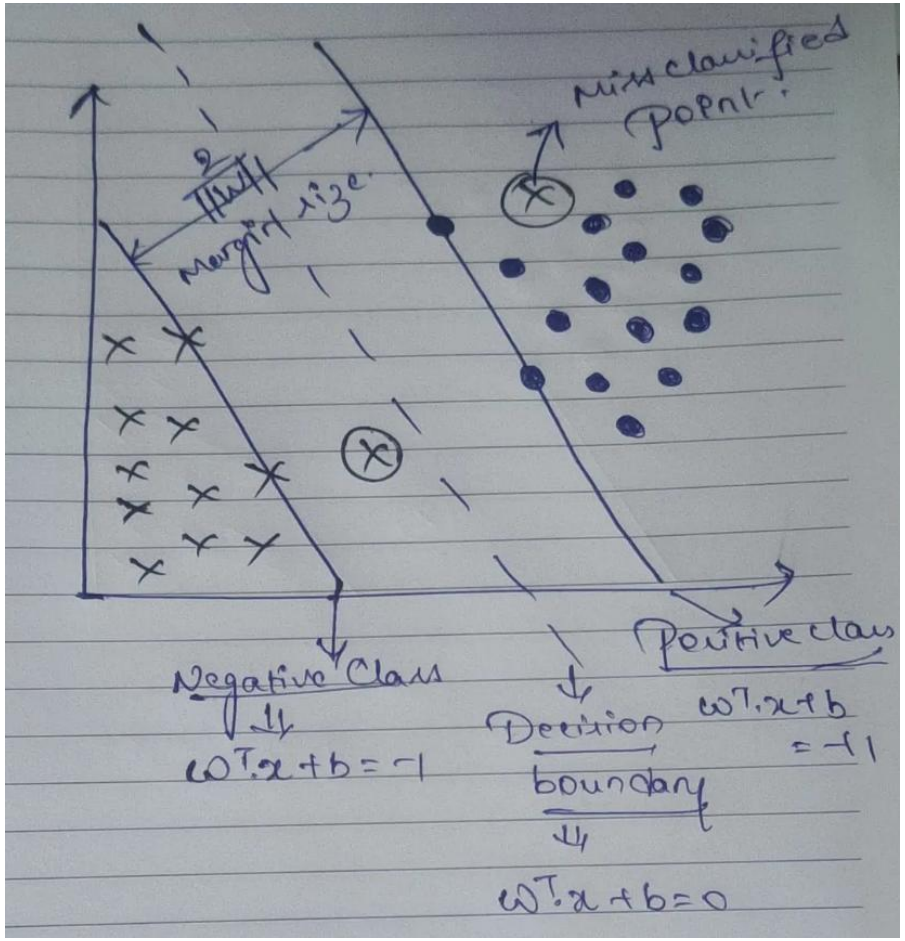
Soft margin classifier

- Hard margin can’t handle non-linearly separable problems. And also it forces these constraints on the data to get linearly separated.
- What if data is NOT linearly separable?
- It doesn’t “allow” misclassification — it **fails to find a solution**.

- Soft margin classifier allows misclassifications making it handle the non-linear data.

Soft margin classifier

- There is a data point that actually belongs to the negative class but is misclassified as the positive class
- There is another data point that is correctly classified as the negative class but it is inside the margin.
- So for both the data points there is actually some error. That error needs to be minimized.
- *This soft margin classifier deals with maximizing the margin as well as Minimizing the misclassification error.*



Hinge loss function

- This hinge loss function tells about how much loss or error has been incurred for each data point. For correctly classified data points the hinge loss will always be zero and for misclassified points, the error will be greater than zero.

from the decision rule,

w.k.t,

$$y_i(w^T.x+b) \geq +1$$

$\Downarrow$

This is true for any value of y.

from this:

$$\Rightarrow 1-y_i(w^T.x+b) \leq 0$$

Now the hinge loss will be

$$\boxed{L(y, f(x)) = \max(0, 1-y*f(x))}$$

$$\boxed{f(x) = w^T x+b)}$$

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Hinge loss for 4 cases of data points

- For a correctly classified data point that lies in the positive class [y=1]:
  max(0,1-y*f(x)) = max(0,1–1*(>=1)) = 0.
- For a correctly classified data point that lies in the negative class [y=-1]:
  max(0,1-y*f(x)) = max(0,1-(-1)*(<=-1)) = max(0,1*(<=-1)) = 0
- For a negative data point that lies in the positive class [y = -1] :

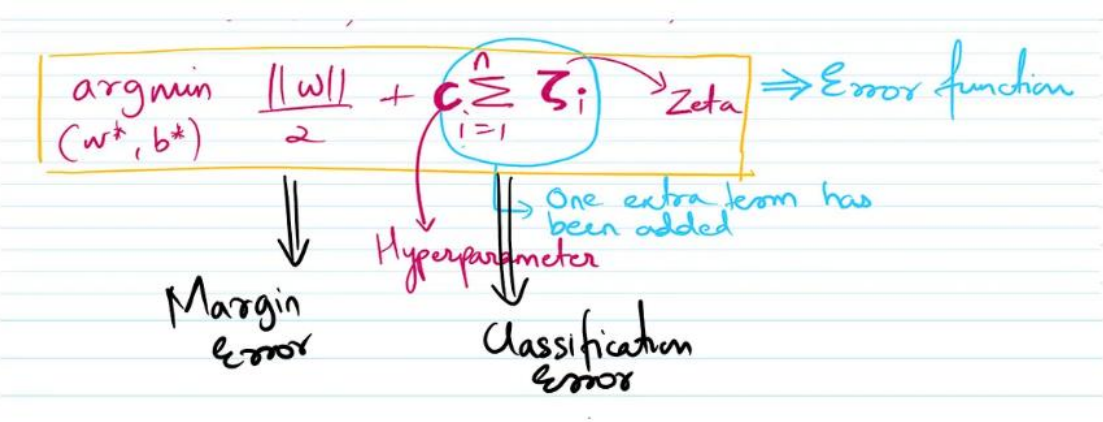
  max(0,1-y*f(x)) = max(0,1-(-1)*(>1)) = max(0,>1) = >1
- For a negative data point that is correctly classified but it's inside the margin [y=-1]:

  max(0,1-(-1)*([-1 to 0])) = max(0,[-1 to 0]) = [-1 to 0]

Hinge loss - Inferences

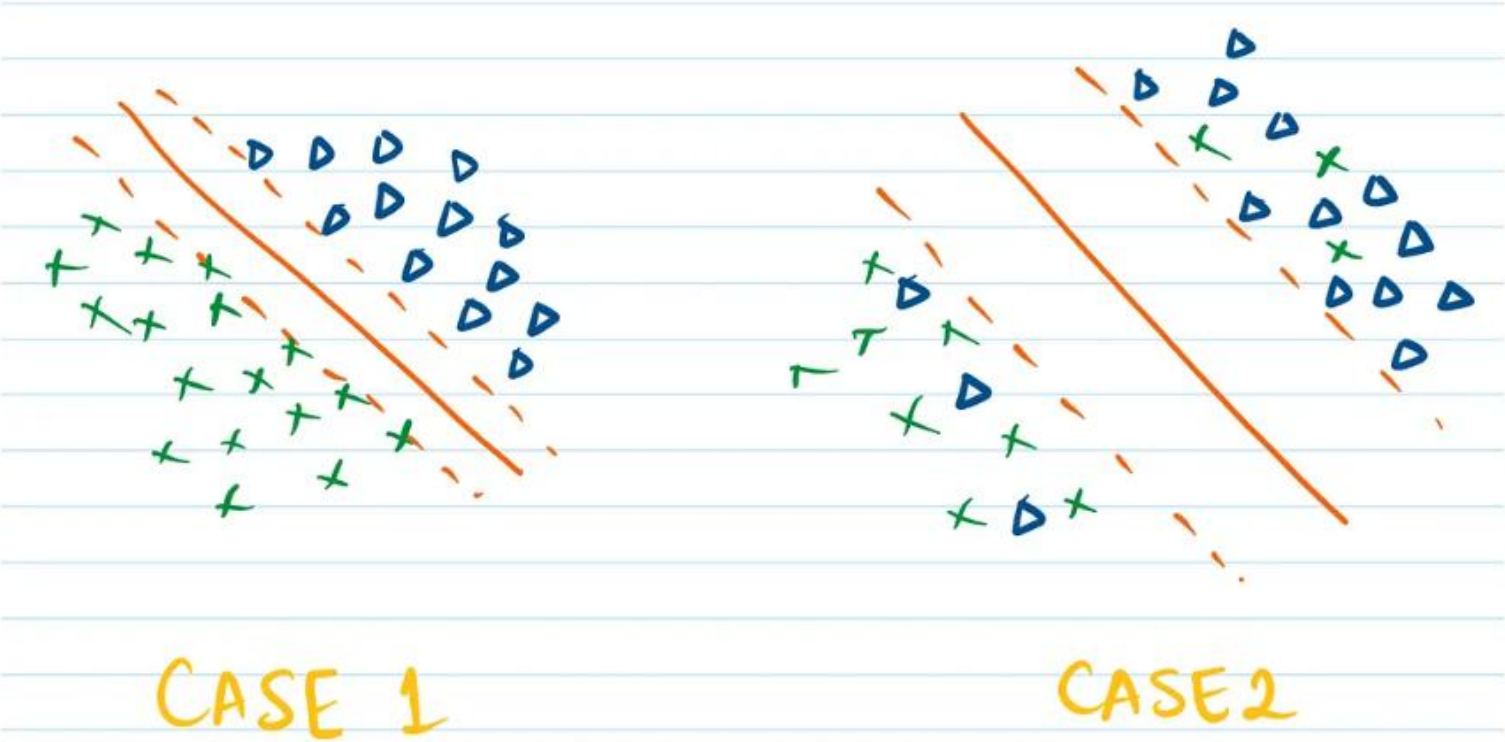
- For correctly classified points the loss will always be zero.
- The negative point which is classified as the positive has a loss greater than 1.
- The correctly classified negative point which is inside the margin has a loss ranging between -1 to 0.
- *Now this loss, which is known to be the* ***Misclassification error*** *should be minimized.*

Soft margin – SVM error



The zeta term is the Hinge Loss

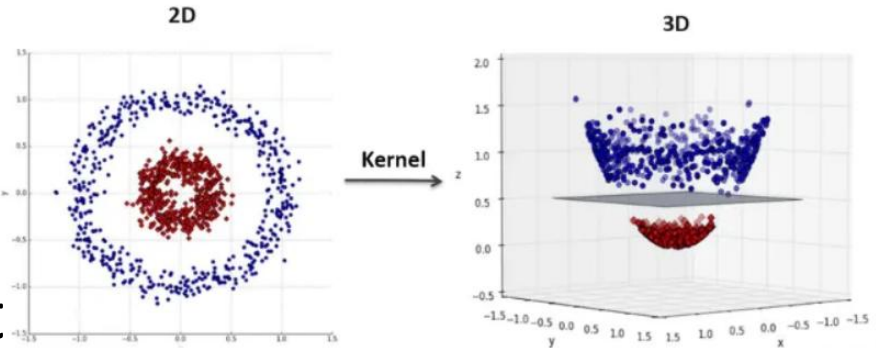
SVM Error is a combination of two things: **Margin Error + Classification Error**

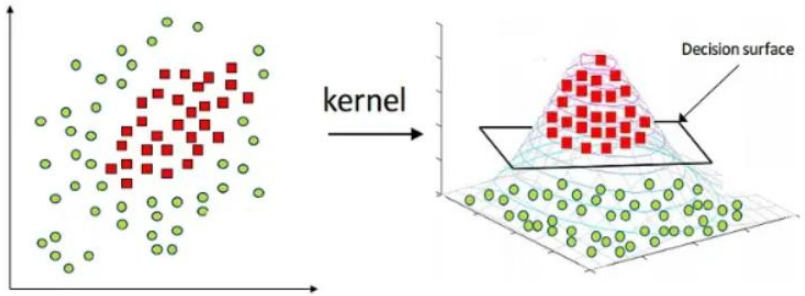


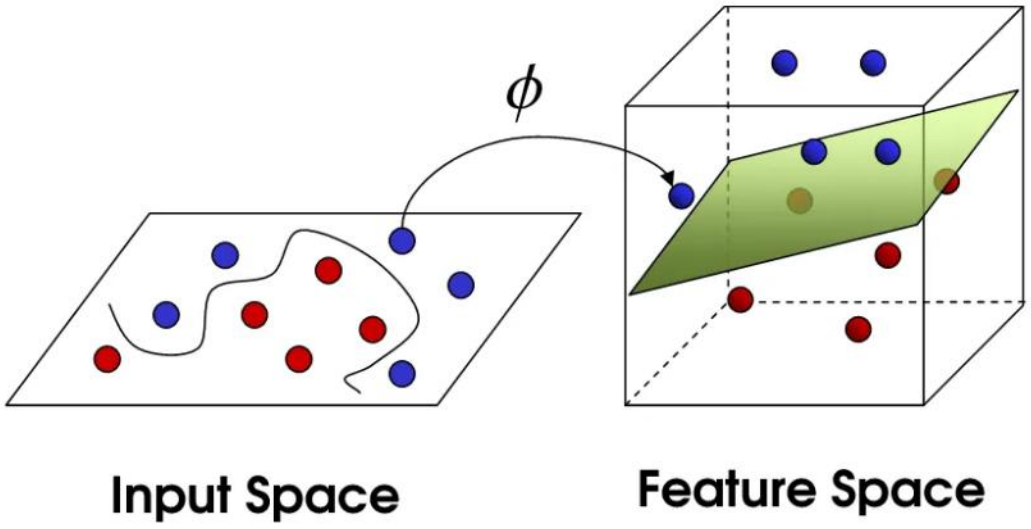
- Case 1 : If we increase the value of C which is a hyperparameter to a very big value, it means we are giving more importance to classification error and less importance to margin error, which means the margin will be very small.
- Case 2 : If we decrease the value of C, then more importance is given to margin error which means the margin will increase but number of classification will also increase

SVM kernels

- The primary goal of an SVM is to find a hyperplane that best separates different classes of data points.
- However, in many real-world scenarios, the data is not linearly separable in the original feature space.
- Kernels help by implicitly mapping the original feature space into a higher-dimensional space where the data might be more easily separable.







The original feature space is the space defined by the input features of the data. Each dimension in this space represents one feature.
For example, if your data has two features (say, height and weight), the original feature space will be two-dimensional.

SVM Kernels

- **Non-Linearity Handling:** Kernels allow SVMs to handle non-linearly separable data by transforming the feature space. This is achieved without explicitly performing the transformation, which can be computationally expensive.
- **Flexibility:** Different kernels can be used depending on the nature of the data and the problem at hand, allowing SVMs to adapt to a variety of tasks.
- **Feature Extraction:** Kernels can implicitly perform feature extraction by projecting data into a space where it becomes linearly separable.
- When data is not linearly separable in the original feature space, SVM uses a method called the **kernel trick** to map the data to a higher-dimensional feature space.

- **Higher-Dimensional Feature Space**: By applying a kernel function, the data is transformed into a new, higher-dimensional space where the data may become linearly separable. In this new feature space, SVM can find a linear hyperplane that effectively separates the classes, even though the data appeared non-linear in the original space.
- The key idea of SVMs is that we don’t need to explicitly compute the mapping to the higher-dimensional feature space. Instead, the **kernel function** computes the similarity between data points in the higher-dimensional space without having to directly compute the coordinates of each point in that space. This allows SVMs to handle complex, non-linear relationships between features while maintaining computational efficiency

Commonly Used SVM Kernels

- Common kernel functions for SVM
  - linear $k(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1 \cdot \mathbf{x}_2$
  - polynomial $k(\mathbf{x}_1, \mathbf{x}_2) = (\gamma\, \mathbf{x}_1 \cdot \mathbf{x}_2 + c)^d$
  - Gaussian or radial basis $k(\mathbf{x}_1, \mathbf{x}_2) = \exp\left(-\gamma \|\mathbf{x}_1 - \mathbf{x}_2\|^2\right)$
  - sigmoid $k(\mathbf{x}_1, \mathbf{x}_2) = \tanh(\gamma\, \mathbf{x}_1 \cdot \mathbf{x}_2 + c)$

- **Linear Kernel**
  - The linear kernel is the simplest and most straightforward kernel function.
  - This kernel is used when the data is already linearly separable. It effectively means that no transformation is applied to the data.

- **Polynomial Kernel**
  - The polynomial kernel allows for more complex decision boundaries by adding polynomial features to the data.
  - This kernel can capture interactions between features up to a certain degree.

- **Radial Basis Function (RBF) Kernel**
  - The RBF kernel, also known as the Gaussian kernel, is a popular choice due to its flexibility. It is defined as:
  - This kernel can handle very complex and non-linear relationships.

Choosing the Right Kernel

- Selecting the appropriate kernel for your SVM model depends on several factors:
- **Data Complexity:** For linearly separable data, the linear kernel is sufficient. For more complex data, consider polynomial or RBF kernels.
- **Computational Resources:** RBF and polynomial kernels are computationally more intensive than the linear kernel. Ensure that your computational resources can handle the increased complexity.
- **Model Performance:** Experiment with different kernels and use cross-validation to determine which kernel yields the best performance for your specific problem.